Adaptive behavior

In 1972, British evolutionary biologist John Maynard Smith introduced a theory known as the evolutionarily stable strategy (ESS), that helped explain how behavioral strategies appear by natural selection. Just as factors such as food and temperature can affect animals, so can the behavior of other species. Maynard Smith suggested that an ESS adapts to the behavior of other animals, and cannot be beaten by competing strategies, thus giving animals the best chance to pass on their genes. He argued that only natural selection could upset this balance—hence why an ESS is “stable”—and that these behavior patterns are genetically preprogrammed.

ESS has its roots in game theory: a mathematical way of working out the best strategy in a game. Many examples of how animals behave emerge as being evolutionarily stable strategies, such as territorial behavior and hierarchies. For example, the genetically pre-programmed “rules” of “if resident, fight and defend” or “if visiting, give in and retreat,” which would help animals retain territory, combine to make territorial behavior an ESS.



Behavior arising from conflicts over space and territory might emerge as evolutionarily stable strategies. Fruit bats jostle for the best spots in the trees, with alpha males driving weaker bats down to lower branches.